

Justification: *Add clarity around max allowed value and how to request no (0) delay.*

6.24.6 DC Dimmer Command 2

Table 6.24.6a defines the DG attributes. The signal and parameter attributes are found in Table 6.24.6b.

Table 6.24.6a – DG definition

DG attribute	Value
Name	DC_DIMMER_COMMAND_2
DGN	1FEDBh
Default priority	6
Maximum broadcast gap	N/A
Normal broadcast gap	As needed
Minimum broadcast gap	As needed
Number of frames	1
ACK requirements	NAK, DC_DIMMER_STATUS_3

Table 6.24.6b – Signal and parameter definition

Byte	Bit	Name	Data type	Unit	Value description
0	-	Instance	uint8	-	Instance number the command applies to. Valid = 1 to 250. Set to FFh for group commands.

1	-	Group	uint8	bitmap	see Table 6.22.2a If bit 7 = 1 and bit 6 = 0, it is a node group. Node groups support more than seven groups where multiple groups cannot be selected in one command. This is required to handle the physical grouping of multiple control instances within one node. 10000001 – Node Group 1 10111111 – Node Group 63 11111111 – For non-group commands
2	-	Desired Level (Brightness)	uint8	%	See Table 5.3 230-249 = Scene 1-20 Memory Value 250 selects the Dimmed Memory Value 251 selects the Master Memory Value
3	-	Command	uint8	-	See Table 6.24.6c for a list of possible commands and explanations.
4	-	Delay/Duration	uint8	Sec	Number of seconds to wait before executing command or the number of seconds of duration for the specified command (for duration commands) before reverting to previous state. Max 240 seconds. Additional minute increment values: 241 = 5 min 242 = 6 min ... 250 = 14 min 252 = 16 min (this is the maximum value allowed per Table 3.2.3b) For Delay Commands: A value of 0 indicates no delay. If a value of 255 is sent with no previous value defined, it

					shall be interpreted as no (0) delay. For Duration Commands: A value of 0 indicates a momentary command (instance will revert to previous state after 100 ms (0.1 seconds) if another command is not received). A value of 255 indicates continuous duration.
5	0 to 1	Interlock	uint2	-	00b – no Interlock active 01b – Interlock A 10b – Interlock B A command message with either interlock A or B set will not be activated until an identical message is received from a different source with the opposing interlock set.
6	-	Ramp Time	uint8		0-250 - Full ramp time from current brightness to new brightness in 0.1 second increments (0 to 25.0 seconds)

Table 6.24.6c – Supported Command Descriptions

Command	Lock Support	Description
00 – Set Level (delay)	Yes	Set output level directly to the ‘desired level’
01 – ON (duration)	Yes	Set output on directly to 100%
02 – ON (delay)	Yes	Set output on directly to 100% delayed by the value in ‘Delay’
03 – OFF (delay)	Yes	Set output off directly to 0%. This can be delayed by the value in ‘Delay’ if greater than 0.

04 – Stop	No	If ON, set output directly to 0%. If flash is active, stop the flash and set output to off.
05 – Toggle	Yes	Toggle output between 0% and ‘desired level’.
06 – Memory OFF	Yes	Store current output state to Master Memory Value for the instance and the set directly to 0%.
07 – Save Scene	No	Store the current brightness to the scene memory value of the scene selected by the desired level (230-249).
11 – Ramp Brightness	Yes	Ramp brightness to ‘desired level’.
12 – Ramp Toggle	Yes	Toggle brightness between 0% and ‘desired level’ each time received.
13 – Ramp Up	Yes	Ramp brightness up from current brightness until either at 100% or a ‘Stop’ is received.
14 – Ramp Down	Yes	Ramp brightness down from current brightness until either lowest brightness is reached or a ‘Stop’ is received.
15 – Ramp Up/Down	Yes	Ramp brightness down to the lowest level. Then start ramping up until 100% is reached. This continues until a ‘Stop’ is received.
16 – Ramp Up/Down Toggle	Yes	Ramp brightness up/down to the highest/lowest level and then stop. Direction of ramp is depending of the current brightness before starting ramping.
21 – Lock	-	When received, certain commands for this instance will be ignored until an ‘Unlock’ is received. The commands to be locked (ignored) are device specific. In safety-critical applications, lock should not be the sole method of safety used.
22 – Unlock	-	When received, removes lock condition for instance allowing all other commands to be actuated.

31 – Flash	Yes	Alternately set the output to 0% and 100%. Delay/Duration value sets the flash period. Desired level sets the duty cycle as a percent (1-199) of the duration. For example, a duration of 30 with a brightness of 33% would be on for 10 seconds and then off for 20 seconds. Invalid brightness values (0, 255, etc) will default to 50%.
32 – Flash Momentary	Yes	Flash output alternately to 0% and 100% once or continue only while the command is being repeated at least once every 2 seconds. Otherwise, flashing will stop.